

Adjusting Saxophone and Clarinet Reeds

Peter Spitzer

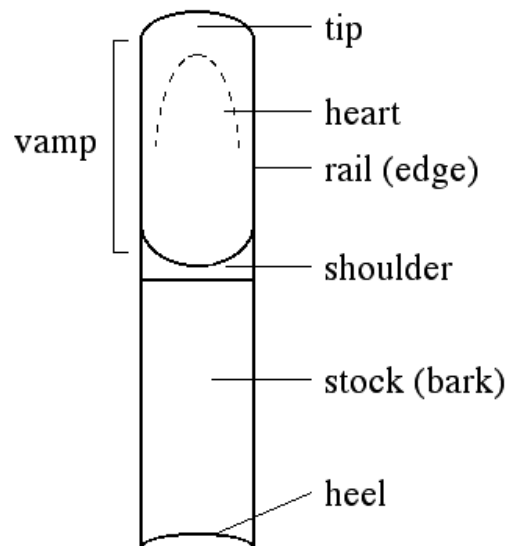
I've been reading some interesting books on reed adjustment: *The Single Reed Adjustment Manual* by Fred Ormand, *Selection, Adjustment and Care of Single Reeds* by Larry Guy, *The Saxophone Reed - The Advanced Art of Adjusting Single Reeds* by Ray Reed, *Perfect A Reed...and Beyond* by Ben Armato, and the *Handbook for Making and Adjusting Single Reeds* by Kalmen Opperman. There are also some good pages on the subject in *The Art of Saxophone Playing* by Larry Teal. My motivation, of course, has been to try to address the eternal problem of how to find a good reed.

Reading these books has been an enlightening experience; the authors all have real expertise. There is much agreement, some disagreement, and each has a somewhat different slant.

What makes a reed "good" or "great" is highly personal. Every player has his or her own combination of embouchure, mouthpiece, and instrument. Most importantly, each player has a personal concept of a desirable sound. There is no way that a commercial reed maker could satisfy every player's individual needs. Also, quality control at the factory could be better. Thus, unless you are willing to throw out a lot of reeds in the search for a "good" one, adjustment is a worthwhile skill to develop.

This article has four parts:

- A practical "how-to" summary of reed adjustment, based on my own experience, combined with information I've picked up from these books;
- Some notes about reed-adjusting lessons with Joe Allard, from my friend Robert Kahn;
- A brief review of each of the books; and
- Some observations on areas of agreement and disagreement.



Basic reed adjustment

This section really only “scratches the surface” (so to speak). I have had to leave out a large amount of information, in the effort to be concise.

Our topics here are: Minimum Tools, Reed Selection, The Breaking-in Process, Vamp and Tip Adjustment, Warps, Weather, and Continuing Adjustment.

Minimum tools:

- Reed knife - I like a beveled-edge knife (they come in left- or right-handed). Keep the knife very sharp.
- Sharpening stone - should have a coarser and a finer side.
- Sandpaper - #400 “wet-or-dry.” You may find it useful to also have on hand #600 (for fine work, like near the tip) and #320 (for rougher work, like near the shoulder).
- A flat surface - a small glass or plastic plaque, 4” x 6” more or less.
- A reed clipper (for clarinet, alto, tenor, or bari).

Super-minimum tools (suggested for beginners): One sheet of #400 wet-or-dry sandpaper and a very flat surface. Using scissors, cut the sheet into quarters. Save one of the quarters for sanding the table (bottom) of the reed, when necessary. Cut one of the other quarters into small strips about one quarter inch wide and 1 1/2 inches long; cut one end of each strip diagonally, into a point. Use these strips instead of scraping with a reed knife. Make a lot of them; they wear out quickly (credit to Larry Guy for the strips idea).

Reed Selection:

Some reeds have more potential than others. Start with a box of new reeds, or a pile of your old barely-played-but-rejected reeds. Choose a few that meet the following criteria:

- Hold the reed up to a strong light. The opaque part (heart) should be in more or less an inverted “U” or “V” shape. The tip and sides should show the light, and be more or less symmetrical between the left and right sides. The “fibers” (xylem/phloem bundles) should also be symmetrical, and finer is better. These fibers are the hardest part of the wood, along with the bark.
- The cut at the shoulder (middle of the reed, between the stock and the blade) should be symmetrical.
- The heel (at the bottom of the reed, the end of the stock) should be symmetrical, showing edges (sides) of equal height. The upper “bark” side should show an even curve, highest point right in the middle.

If you start with 10 good candidates, you may, with skillful adjustment and good luck, end up with 8 reeds that are at least playable, with one or two of them superior.

Different brands and models of reeds have different dimensions, and may be made from cane that has been chosen for different qualities. You will want to find a brand and model that seems to work with your mouthpiece, embouchure, and playing style. If you already know what works for you, you may wish to stick with it. If not, try a few each of some different brands, as an experiment.

Reeds out of the box will change as they are played. A reed can be great one day and awful the next day. The introduction of moisture to a dry reed, and subsequent drying out, will cause swelling and warping. The stress of being played will also contribute to warping. The art of reed adjustment is largely about dealing with these changes.

Breaking-in Process:

It's probably safe to say that most clarinet and sax players, including many or most professionals, just open a box and try reeds, picking the good ones out and discarding the rest - perhaps with some effort at adjustment, but without a "breaking-in" period. However, the authors of these books generally advocate a gradual breaking-in process. Below is a basic summary. If you are in a hurry, or don't completely agree with this logic, you can adapt it to your needs - for example, by shortening the process to just a couple of playing sessions.

The concept is to play the reed for only a few minutes each day for perhaps 4 to 10 days, making very minimal adjustments each time. This way, the reed will only gradually begin to exhibit the warping and swelling pattern inherent in that particular piece of wood, as you simultaneously try to improve it. In addition, the reed will gradually "learn" to conform to the shape of the mouthpiece facing. At the same time, the surface will become sealed somewhat, and the reed will become less prone to change. At the end of this process, you will have a more stable reed.

At each playing session, first moisten the reed (with clean water or saliva - players have differing opinions), then play for 3-5 minutes, evaluating what adjustments might be necessary. When you think you know what needs to be done, try your adjustment, play-testing after each small knife stroke, sanding, or clip.

For the first few days, dry the reeds flat side up when you are done working on them. After perhaps 3 sessions, put them away after each session in reed holders designed to keep the bottom flat (a flat piece of plastic with a rubber band will work fine).

Vamp and Tip Adjustment:

Adjustments should be gradual. In the first session, concentrate on the part of the vamp (upper surface of the reed blade) nearest to the shoulder. If your low notes are stuffy, scrape here lightly, trying to preserve balance, avoiding the middle of the reed. If the entire reed seems hard, use a small piece of #400 sandpaper to lightly sand the entire vamp, starting at the shoulder, stopping short of the tip. Don't try to perfect the reed at this point; it is too early. Be satisfied with a small improvement, and put the reed away for the day. Do this with each reed in your batch.

In the next several playing sessions, concentrate on balancing the reed. Use the reed knife to very lightly scrape the lower and middle vamp, as necessary. The object is to equalize the vibrations of the left and right sides. Here are three ways to check balance:

- Turn the mouthpiece clockwise, so that your embouchure only controls the right side, with the left side of the reed free. Blow an open C# (sax) or G (clarinet) - then turn the mouthpiece the other way, to free the right side. If the "free" side (L or R) seems stuffy compared to the other, some wood should be removed from the stuffy side. I've had some good results with this method.
- Hold the reed up to the light, to look for symmetry. An asymmetrically darker area, or an area with thicker fibers, may have to be thinned slightly (scrape lightly with knife or use a thin strip of #400 sandpaper).
- The L-R balance of the tip can be checked by using your forefinger to flex each corner, in turn. You are looking for equal flex. If one side seems stiffer, use a thin strip of sandpaper (#400 or #600) to lighten the inner part of the flexing tip, where it meets the heart. Avoid sanding or scraping the tip edge itself. Hold off on tip adjustment until later sessions.

In balancing, remember that the "fibers" are the hardest part of the wood. If one side of the reed seems stiffer, and shows heavier fibers, some light sanding or scraping of those fibers might be helpful.

Scrape or sand in the most minimal way possible, then play-test the reed. Once wood is removed, it can never be replaced. Again: be satisfied with gradual improvements.

The smaller the reed, the more minimal your scrapes should be: A small scrape on a clarinet reed is much larger, proportionally, than the same scrape on a tenor sax reed.

After several days, when the reed is generally balanced and is fairly stabilized, work on the tip, if necessary. This is the time to use the tip “flex” test described above.

An important part of tone adjustment is the amount of high overtones in the sound. The high overtones are largely determined by the tip of the reed. If a reed is too bright (or if it is generally too soft), the tip may be clipped, using your reed clipper. Take only a tiny amount off, and play the reed to check the result. A very small clip can make a significant difference. Don't clip the reed in the first few playing sessions, even if the reed seems to play soft. When you do clip it, it's a good idea to leave it a bit soft. The next day, you will often find that it has changed in a “harder” direction.

After clipping, you may want to use a piece of #600 paper to round the corners of the reed to match the tip of your mouthpiece. Corners that stick out can cause an excessively bright, somewhat raucous vibration. Play it first, though, to see if this is really necessary. Hold the sandpaper in one hand (hold one end, let it flop), the reed in the other, and shape the reed tip by pulling it over the paper.

If the reed lacks high overtones, use a small strip of #600 sandpaper (or a delicate stroke of the knife) to lightly thin just below the tip, to the left and/or right of center. Try for balance. Work in small increments, checking your result by blowing a few notes after each couple of strokes.

Although highs are associated with the tip, sometimes lightening the middle and lower sides of the blade can free up “stuffy” highs.

Generally speaking, response of low notes is adjusted at the sides of the bottom of the vamp, near the shoulder. Middle notes are adjusted on the sides of the middle of the blade, and highs towards the tip. If possible, try to avoid sanding or scraping the center (heart) or the extreme tip of the reed.

If you overdo the scraping or sanding, and the reed becomes too soft, you can clip the tip, then try to reshape and rebalance the reed.

I should mention here that you learn by making mistakes. I've ruined plenty of reeds in trying to learn how to work them, but I've learned a lot. I still have a lot to learn, and I expect to ruin many more reeds in the effort.

Warps:

Reeds will warp in the normal course of being moistened/played/dried. Warping can occur on either the top (vamp) or the bottom (table) surface. Warping on the bottom of the reed can spoil the seal of the stock with the mouthpiece, causing uneven vibration and possibly air leaks. To combat this, run the flat blade of the reed knife lightly over the reed table to flatten it, removing only a small amount of wood dust. Do this each day for the first 2 or 3 days of the breaking-in process (after a few minutes of playing), and later on as necessary.

If a more severe warp develops, the bottom may be flattened by rubbing it lightly over a quarter sheet of #400 sandpaper that has been placed rough side up on a very flat surface (keep the tip off the sandpaper). I try to avoid this if possible, as removing wood from the bottom can affect the reed's tone adversely. If you do sand the bottom, use a light touch.

Some musicians advocate flattening and sealing the bottom by rubbing it over very fine sandpaper, or on the paper side of the sandpaper sheet, or on plain paper. I don't do this, as I think it makes the reed's tone thinner and brighter.

Warping on the top surface will generally be removed in the ongoing balancing process.

If the tip appears warped or “crinkled,” don't worry about it! The waviness will come out in the course of playing. You can get it started straightening out by holding the wavy tip flat with your thumb, against the flat table of the mouthpiece, for 30 seconds or so.

Weather:

High humidity can cause drastic changes in reeds. They can warp, become heavy, and respond poorly. Don't adjust the reed much, or at all, on a rainy day. When the weather dries out, the problems may largely disappear. If you try to make a perfect reed on a wet day, you may later find that you have drastically over-cut the reed. Very dry weather can cause unwanted changes, too (I don't have that problem much here in the San Francisco area).

Continuing Adjustment:

Even if you think you are finished with the adjusting, and have an excellent reed, it will change over time, and may need further balancing or clipping.

With use, both sides of the reed will accumulate a thin layer of deposits (gunk) from evaporated saliva, dust, etc., which will stifle vibration and deaden the tone. On the top of the blade, this may be removed with very light strokes of the reed knife, or with your fingernail. To clean the bottom, run the flat blade of the reed knife lightly over the entire surface, starting at the heel end, stopping just short of the tip.

Don't forget to brush your teeth before playing, or at least rinse your mouth. Your reeds will last longer. After each gig or practice session, you should, if possible, rinse off the reed with clean water and wipe it dry.

Reed placement can help performance to some degree. If your reed blows hard, try moving it down on the mouthpiece so that the tip of the reed barely overlaps the tip rail of the mouthpiece - it will blow just a little easier. If the reed blows soft, try moving it up so that the reed tip covers all of the tip rail; it will blow a bit stiffer. This can help in last-minute adjustments before a performance, or can serve as a test to see if the reed should be either scraped/sanded, or clipped.

Ligature placement can also affect performance. Larry Guy: "sometimes lowering the ligature slightly frees up a stuffy reed, while raising can help center the sound of a wild one." I should add that different ligature designs will affect the tonal quality. It's worth the effort to try different ligatures, to see what works for you. Generally, the greater the amount of material in the ligature, the darker the sound, and vice versa.

Eventually, after maybe 10 hours of playing, even the best reed will lose its resiliency. Sometimes you can coax a little more use out of it by clipping the tip. You can also try soaking it for a few minutes in (1.) 3% hydrogen peroxide or (2.) Efferdent, then washing it off. But at some point you will have to decide that the reed's life is over, and toss it out.

Robert Kahn On His Lessons With Joe Allard

Reed adjusting? Never saw a book on it worth beans, but that's mostly what I did with Joe Allard the 2nd year I studied with him.

Joe was really into working on reeds.

Joe recommended hard reeds because there was more wood to work with.

He recommended the Bhosys reed knife, which I got at Manny's. It's great. You just sort of roll it away from you to shave a leetle bit of wood at a time.

Joe said not to mess with the heart of the reed – just the sides, from the rails in, maybe a quarter of the way in, so when you were done, you'd still have the middle half untouched.

Joe would put the reed on and blow it – any note – open C# - then twist the mouthpiece a bit, so that one side of the reed would be sealed, and only the other side would vibrate. Then he'd do that to the other side and blow it. One or both sides would sound muffled. That's the side you start taking some wood off of. The extra wood kept the reed from vibrating. He'd start down near the base of the reed and work up, at a bit of an angle, like this on the right side: / , sort of – staying away from the heart. Stay away from the tip too, he said. It's already thin there.

He'd take a little wood off one side, maybe both, and then repeat the blowing thing – blow it normal, blow it with one side sealed, then the other one. Once one side sounds as bright as the normal way, leave it alone. Work on the muffled side until it brightens up. Then you're done.

But as you know, reeds change. You might have to do it again later.

He'd also use the flat side of the reed knife to clean off the flat part of the reed – wet the reed and scrape it on the flat side of the knife to get the accumulated gunk off it.

And if you have to clip a reed, Joe always used the next biggest reed clipper to do it – a tenor reed clipper to clip an alto, a bari reed clipper to clip a tenor reed. I think he used an alto clipper on a clarinet, but I ain't sure about that.

He'd also unscrew the moving parts of the reed clipper and throw them away – the part you snap on to hold the reed, and the screw thing that moves it up and down. He'd just put the reed where he wanted it, hold it there and clip it, and then repeat the thing with the reed knife.

I can't tell you how much I dig my Bhosys reed knife. Ask around if you don't have one. It's great because it doesn't dig into the reed, just scrapes off the surface.

Holding the reed up to the light, you can see where it's thick and where it's thin.

Generally, when you find the side with too much wood on it, you'll see it's dark, irregular. That's the part you work on. If there's a light spot on the muffled side, I'd stay away from it until you work on the dark parts.

No breaking in - but keep it clean. Joe didn't seal his reeds in any way.

Note: You can read more about Joe Allard's views on reeds at www.joeallard.org.

Book Reviews

These books are all available from Amazon or from Van Cott Information Services (www.vcisinc.com).

The Single Reed Adjustment Manual by Fred Ormand. 84 pp.

This is the most complete of the books reviewed here. If you are going to buy and read just one of these books, this is the one I'd recommend. It is clearly written, well-organized, and more or less in line with my own experience. Ormand's focus is on clarinet reeds, but all the principles are of course applicable to saxophone also. The book includes chapters on all pertinent topics, and a bibliography.

A few of his views: Advocates a breaking-in process of "5 to 10 minutes, 2 or 3 times a day at most," but doesn't say how many days. Does, however, say, "The harder the cane, the more closely the player should bring it to the finished stage the first day." Believes in using water to moisten the reed, rather than saliva. Says that aging reeds (before adjusting) for between 1-8 years can improve the cane.

Ormand includes an explanation of the difference between "French Cut" reeds and "American Cut" reeds. French Cut reeds (Vandoren Blue Box, Vandoren Java Red Box, Rico Royal, Rico Grand Concert) are stronger at the spine and lighter at the rails, and have a strip of bark removed at the shoulder. American Cut reeds (Rico, LaVoz, Vandoren

Java Green Box) have a broader heart with “somewhat uniform thickness across the width of the reed,” and have the bark left on the shoulder, forming a “wing” of bark on each side. Ormand covers the advantages and disadvantages of each type.

Selection, Adjustment and Care of Single Reeds by Larry Guy. Rivernote Press. 56 pp.

The heart of this book is a guide to a 10-day breaking-in routine, with additional chapters on tools, continued adjustment, and weather. Many excellent points of advice are included in the course of the 10-day description (selection, balancing, etc.). Guy also includes a two-page summary of the 10-day routine, and a useful two-page “troubleshooting guide.” The text is sprinkled with one-line quotes from eminent clarinetists (Gigliotti, Marcellus, etc). His 10-day process involves playing the reed for short periods, beginning with 2-3 minutes daily, increasing to 10 minutes at day 10, while making gradual adjustments.

Some of his views: Says that his own breaking-in routine usually takes longer - up to 4 weeks. Recommends water to moisten reed, after the first few sessions. Advocates rubbing down the reed to seal it, and sanding the heel to seal it. Says that aging reeds for up to 3-6 years can be beneficial.

The Saxophone Reed - The Advanced Art of Adjusting Single Reeds by Ray Reed. Infinity Publishing Co. 195 pp.

This is in some ways the most detailed book considered here, with much insight not found elsewhere. Ray Reed has made a great effort at understanding all the forces involved in reed performance, and how to bring them into balance. He uses some jargon of his own (I think) devising - e.g., primary warp, stress riser, illusion of the reed tip, dynamic balancing, resonance-reversal point. Although he defines these terms for the reader, it makes re-reading necessary. This book is perhaps best suited to those who already have some experience with the more “mainstream” terms and procedures.

Much of the book is concerned with identifying and dealing with characteristic warping patterns.

Reed describes both “fully balanced” and “quick-cut” procedures. Full balancing involves more extensive wood removal, up to 10-12 breaking-in sessions, and preferably the use of a micrometer. The quick-cut method involves more moderate adjustment, as few as 2 breaking-in sessions, and “empirical” evaluation rather than by micrometer.

Because his “fully balanced” method involves so much sanding and cutting, Reed suggests using tenor or baritone reeds as blanks for creating alto reeds (and alto blanks, for soprano). He says, however, that tenor or baritone sax players don’t need to start with a larger reed, as the regular size will usually have enough wood to withstand the “fully balanced” process.

This book is a must-read for the serious student of reed adjustment. As always, you don’t need to take all of his advice, or completely agree with his opinions.

Perfect A Reed ...and Beyond by Ben Armato. 43 pp.

This is a great little book, mixing lots of good advice with some highly personal views. The writing is not always 100% clear, and some of his ideas may seem questionable, but who can ignore advice from a guy who played clarinet with the Metropolitan Opera Orchestra for 35 years?

Armato does not believe in aging reeds. He says that it takes 10 playing hours for the reed to “reach its highest level of performance,” and 10 more hours to deteriorate. Some of his very good advice: “Final reed selection or adjustments should be undertaken at the place of performance,” and “Perfection is only one knife stroke away from disaster.” He does not discuss a breaking-in period.

A few ideas seem odd. He specifies 4 areas for tip adjustment - L corner for high notes, L middle tip for staccato, R middle tip for attacks, R tip for resonance (why exactly should these qualities respond better to adjusting in L vs. R areas?). Another odd idea: cutting a groove across the reed’s table, from one side to the other, under the middle of

the vamp, to enhance the sound. Another: cutting grooves in the mouthpiece rails directly under the spot where the reed leaves the curve of the mouthpiece (what about air leaks?). But then, perhaps I'm missing something - after all, Armato played with the Met for 35 years!

You should check this book out, and see what you think. By the way, Armato is the inventor of the [Reed Wizard](#).

Handbook for Making and Adjusting Single Reeds by Kalmen Opperman. M. Baron Company, Inc. 44 pp.

Although this book primarily deals with making reeds from tube cane, there are 7 pages at the end that are quite valuable to those who are adjusting commercial reeds. This book first appeared in 1956, making it one of our earlier sources. Opperman's writing is clear and succinct. His advice on many points matches the general consensus - e.g., make gradual adjustments, reeds that seem soft at first may stiffen up with use, adjust low notes on lower blade first, etc.

Some points more specific to this book: Before first playing, soak reeds with saliva "a few moments each day, for 3-4 days." Vamp should match length of mouthpiece window. "Reeds made of dense cane usually have a brittle quality."

The Art of Saxophone Playing by Larry Teal. Summy-Birchard Music. 111 pp.

This excellent book covers all aspects of saxophone playing. A 9-page chapter on reeds covers reed mechanics, selection, and adjusting. He recommends a breaking-in period, but doesn't say for how long - just "until you feel that the cane has stabilized its character."

Teal likes using Dutch rush to balance the sides and tip, and minimizes knife use. He advocates sealing the vamp by massaging it with a spoon or the side of a plastic pen. Others may have different views about these two points; pretty much everything else Teal says reflects the general consensus (e.g., reed selection criteria, the 3 balance tests, avoiding changing the heart and tip if possible). He includes a schematic of the reed showing which areas to adjust in order to remedy various problems ("buzzy or edgy," "thin high register," "heavy low register," etc.). The book is well-written, and is a saxophone classic.

Areas of Agreement/Disagreement

Agreement:

Most of these authors seem to agree on some breaking-in period, though the preferred length of time differs. Most agree on limited playing at the first several sessions. Apparently Joe Allard did not teach any breaking in.

All seem to agree on areas of the vamp that affect highs/mids/lows.

All agree on reed selection criteria: lighter fibers, even fibers, balanced heel and shoulders, etc.

Disagreement:

- **Expected life** - Different books say to expect anywhere from 10 to 40 hours of playing. I'd say that a good reed is at its peak for maybe 10 hours, at best. Opperman actually claims up to 100 hours for hand-made reeds (in this article, we are not considering hand-made reeds, but rather the adjustment of commercial reeds).
- **Polishing/sealing** - Some (e.g. Teal) believe in being quite proactive in sealing the vamp and/or table; others believe that this is either unnecessary or counterproductive.

- **Tools** - The tools I have listed above are fairly minimal. You will find disagreement in preferred types of knife, whether to use Dutch rush, use of files, and whether to use a micrometer.
- **Sanding the bottom** - This will take out a table warp, but some believe (as I do) that excess sanding is destructive to tone. Ray Reed seems to suggest taking quite a bit of wood off.
- **Aging reeds** - Is it beneficial to age them (Ormand, Guy) up to 8 years, or pointless (Reed, Armato)?
- **Starting with a harder reed** - For example, see Joe Allard's comment, above. As I understand it, harder does not actually mean thicker, though. It just means that the reed is made of denser wood. I agree with Opperman that "Reeds made of dense cane usually have a brittle quality." Personally, I'd use my normal strength or a half size harder, not more.
- **Working on reeds dry vs. wet** - Most sources advocate playing on a reed for a few minutes before evaluating and adjusting. I would not rule out working on a dry reed, but it makes more sense to work on it after it has been wetted and played for a few minutes, since that is the way it will be when you are actually performing on it.
- **Water vs. saliva** - Some (e.g., Opperman) advocate extensive "soaking" (read: "wetting") with saliva only; others prefer water. Here the logic is unclear - is saliva better, because it is denser and therefore seals the reed? Or is it destructive, as others say (Ormand, Guy), because enzymes in it will break down the structure of the wood? My own feeling is that using water when possible when breaking in keeps the reed cleaner inside and out, and that just through playing, sealing will take place from saliva deposits and/or from normal rubbing when cleaning off the reed. I think wetting the reed with saliva just before putting it on the mouthpiece might make some sense, as the denser saliva might help make a better seal between the flat of the reed and the table of the mouthpiece. Moisten the entire reed bottom, tip to heel.

A Few More Notes

A personal observation about breaking in reeds: There is a point of diminishing returns, in terms of the time spent on reeds that may prove hopeless, and in view of the danger of over-working the reeds. It makes sense to me to employ only 4 or 5 short adjusting/playing sessions, to coax the reed into its characteristic warping, and make a few pre-emptive adjustments, before putting it into service. By then I think I can usually tell which reeds are worth keeping. Adjustments continue, through the useful life of the reed.

You are welcome to disagree or agree with any of these viewpoints, as your experience may dictate. As far as my own opinions go, they are just provisional. It's quite possible that experience will teach me otherwise (even after all these years of playing).

Good luck!